

Customer Churn Analysis: Why Are We Losing Customers?

VortexTech Data Science & Analytics Internship — Week 4 Capstone

Dataset: Telco Customer Churn (7,043 customers of a telecom provider), covering account details, services subscribed, billing information, and whether the customer churned (cancelled service).

Central question: *Why are customers churning, and what could the business do to reduce it?*

Executive Summary

We looked at data from **7,043 customers** of a telecom company to understand why roughly **1 in 4 customers (26.5%)** cancel their service, and what the business could do about it.

The single biggest factor is **contract type**. Customers on flexible, month-to-month plans cancel **15 times more often** than customers locked into two-year contracts. This effect compounds with how new the customer is: nearly **half of customers leave within their first year**, but that risk drops to under 7% once someone has stayed five years or more — loyalty builds quickly if we can keep customers past the danger zone.

We also found that the type of internet service matters a lot. Customers with **fiber optic internet churn more than twice as often** as customers with basic DSL service, which suggests a pricing or reliability problem with the fiber product rather than a demand problem. Separately, customers who don't have **add-on protections like tech support or online security** cancel at close to **three times the rate** of customers who do — these add-ons seem to build stickiness, not just generate extra revenue. Finally, customers who pay by **manual electronic check** churn far more often than those on automatic payment methods, which is often a sign of lower commitment or friction in the billing process.

None of these patterns are about customers being unhappy with the core service everywhere — they're about **specific, fixable friction points**: contract flexibility, the first-year experience, the fiber product, missing add-ons, and manual billing. The recommendations below target exactly these points.

Methodology (Brief)

- **Data source:** Telco Customer Churn dataset, 7,043 customer records, 21 columns (demographics, account info, services subscribed, billing, and churn status).
- **Cleaning:** Converted `TotalCharges` from text to numeric (11 blank values, all brand-new customers with 0 months of tenure, were set to 0). Checked for duplicate records (none found) and confirmed all other columns had no missing values.
- **Analysis approach:** Calculated the overall churn rate, then broke it down across contract type, tenure, internet service type, add-on services, payment method, and billing amount to find where churn concentrates.
- **Tools:** Python (pandas for analysis, matplotlib/seaborn for visualization).

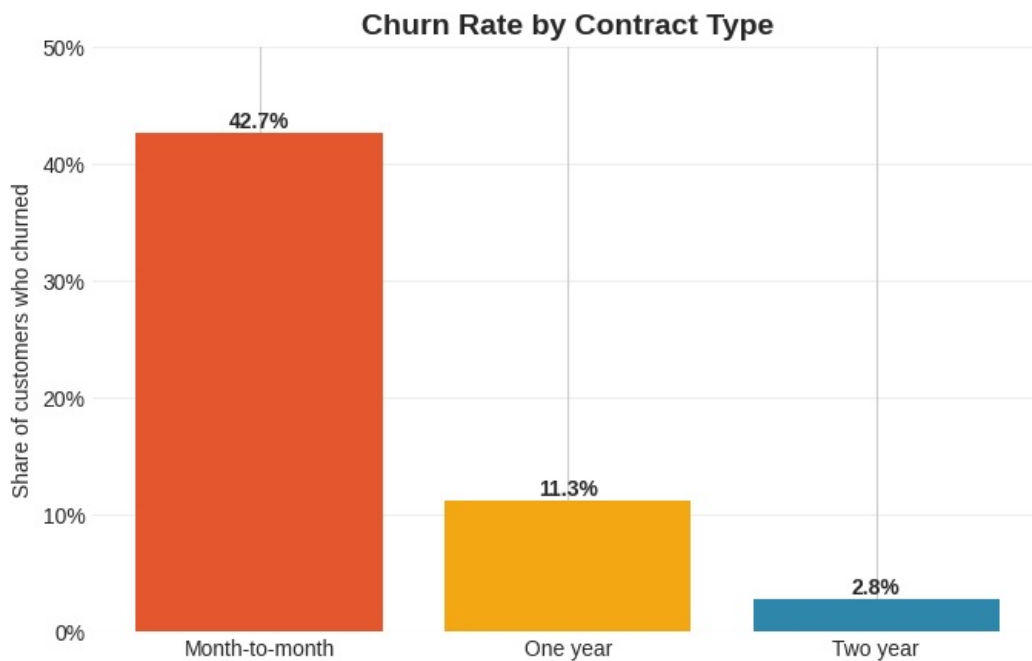
Full cleaning and analysis code is available in the Appendix at the end of this notebook for technical readers.

Dataset ready: 7043 customers, overall churn rate 26.5%

Key Findings

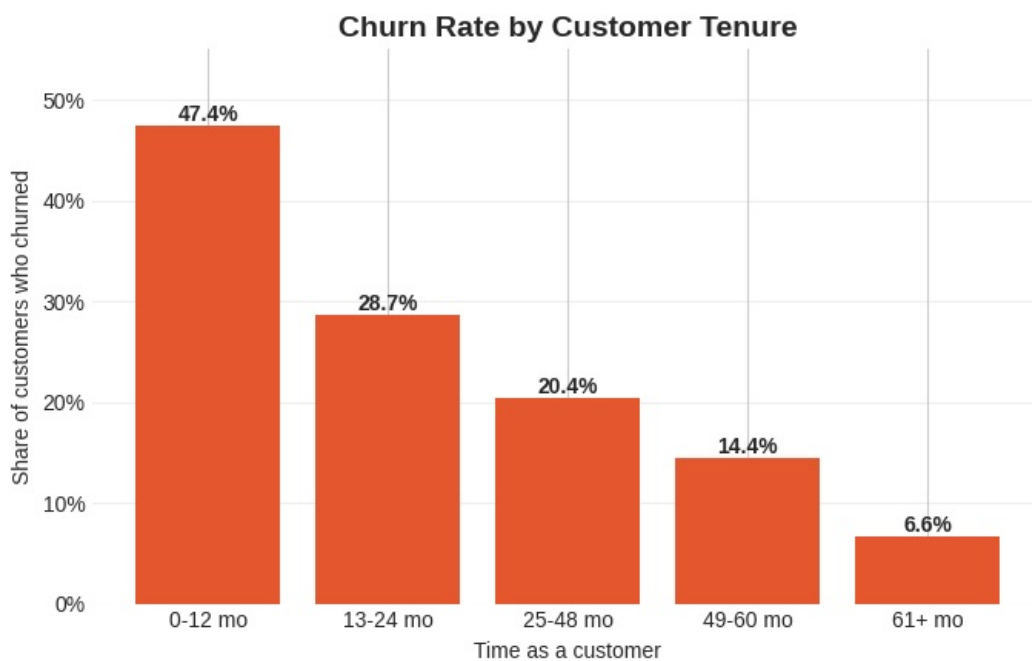
Finding 1 — Contract length is the strongest predictor of churn

Customers on **month-to-month contracts churn at 42.7%**, compared to **11.3%** for one-year contracts and just **2.8%** for two-year contracts. Locking customers into longer commitments is strongly associated with retention.



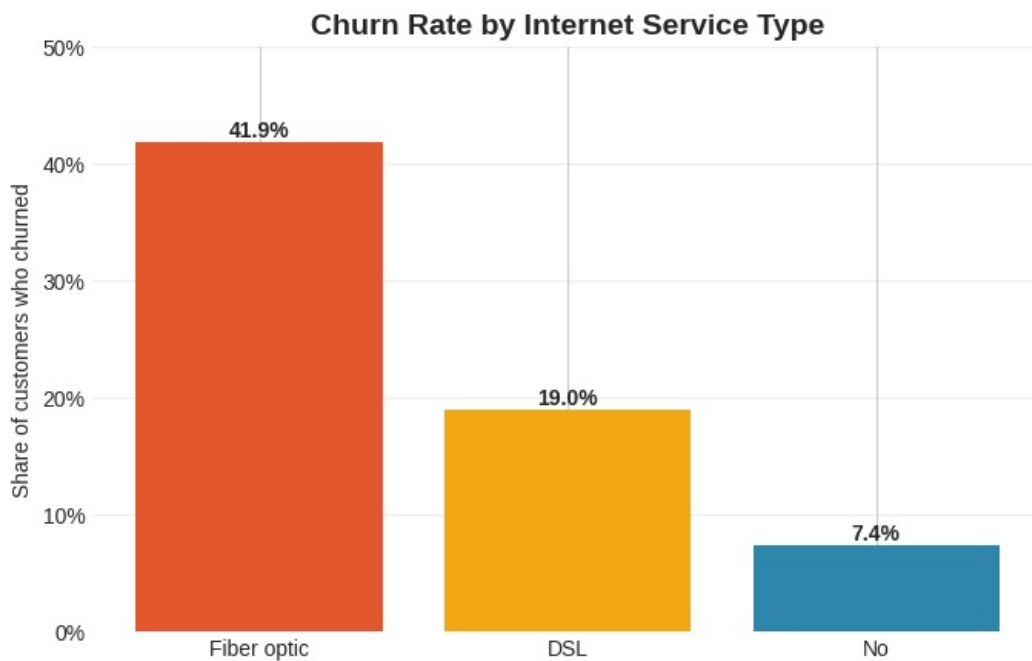
Finding 2 — The first year is the danger zone

Churn risk falls off sharply the longer a customer stays. Customers in their **first 12 months churn at 47.4%**, but that falls to **6.6%** for customers who have been with the company for 5+ years. If a customer can be retained past year one, they are increasingly likely to stay for good.



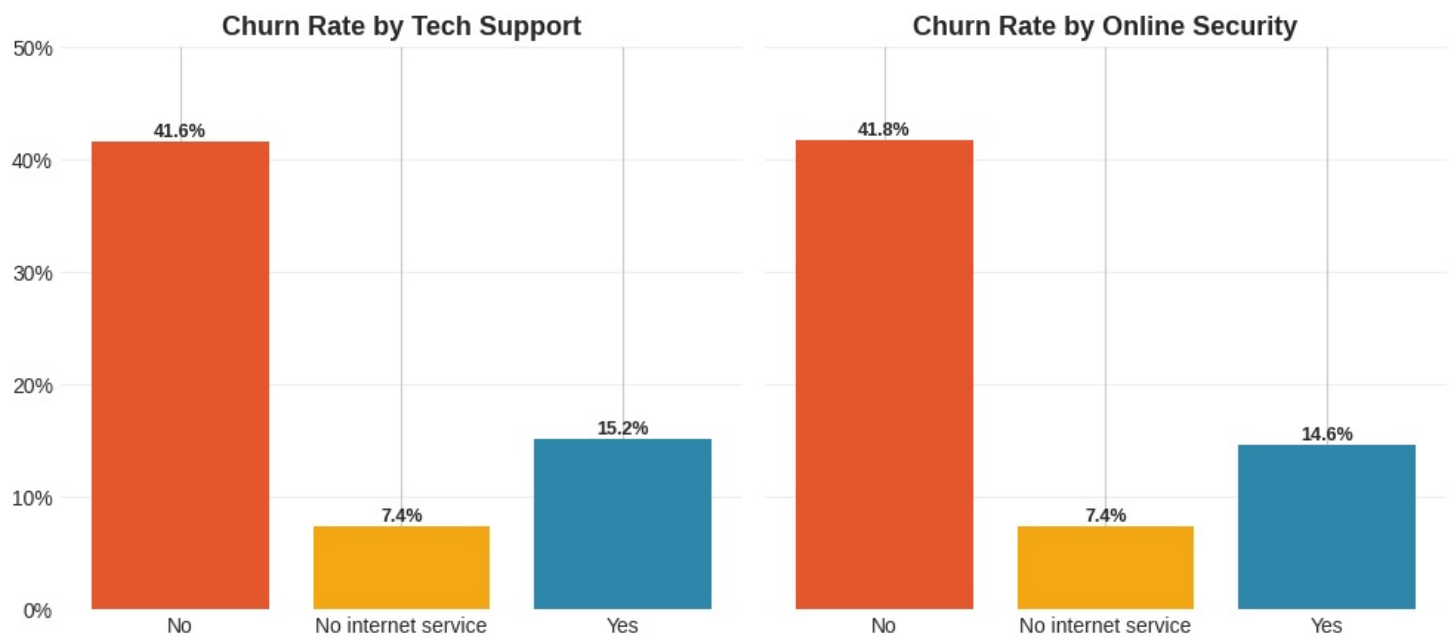
Finding 3 — Fiber optic customers churn more than twice as often as DSL customers

Despite being the premium internet product, **fiber optic customers churn at 41.9%**, versus **19.0%** for DSL and **7.4%** for customers with no internet service at all. This points to a specific problem with the fiber product — likely price or service reliability — rather than a general willingness to churn.



Finding 4 — Customers without protection add-ons churn nearly 3x more often

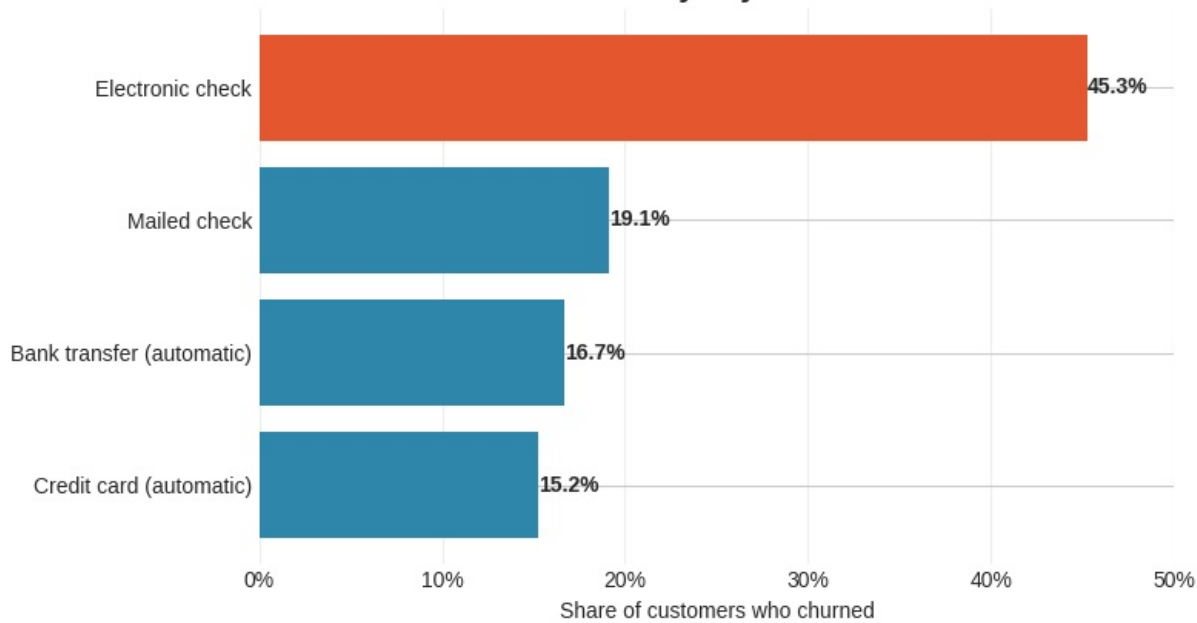
Customers without **Tech Support** churn at **41.6%**, versus **15.2%** for those who have it. The pattern is nearly identical for **Online Security** (41.8% vs 14.6%). These add-ons appear to build loyalty, not just add revenue — likely because they increase how invested a customer feels, and reduce frustration that would otherwise push them to leave.



Finding 5 — Manual payment by electronic check is linked to much higher churn

Customers paying by **electronic check** churn at **45.3%** — far higher than mailed check (19.1%), automatic bank transfer (16.7%), or automatic credit card (15.2%). Manual payment is often a signal of lower commitment, and the extra monthly friction of manually paying a bill may itself contribute to customers deciding to leave.

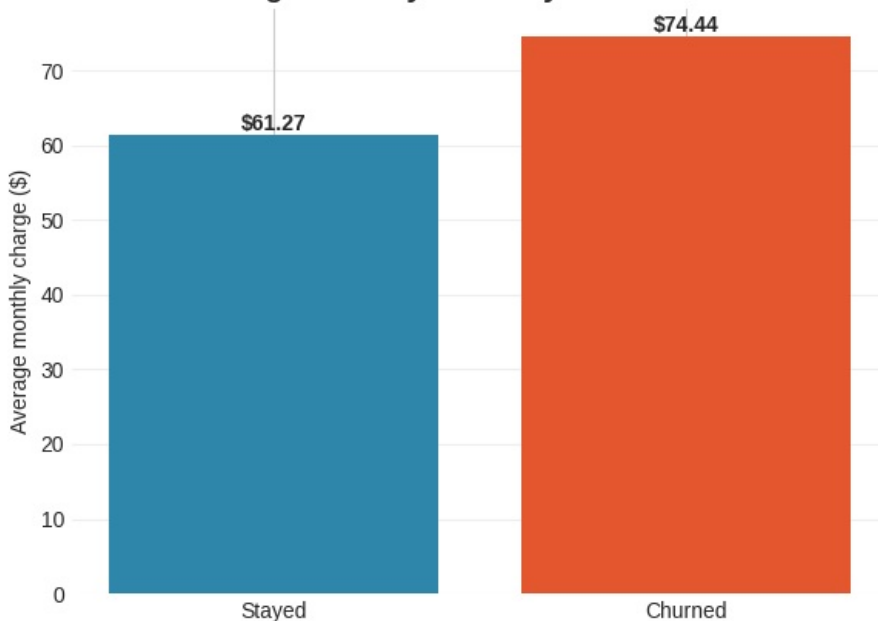
Churn Rate by Payment Method



Finding 6 (bonus) — Customers who churn are paying more, not less

Customers who churned paid **\$74.44/month on average**, notably higher than the **\$61.27/month** paid by customers who stayed. Combined with Finding 3, this suggests some churned customers may feel they're not getting enough value for a higher bill — particularly fiber customers.

Average Monthly Bill: Stayed vs. Churned



Recommendations

1. **Incentivize longer contracts, especially in the first year.** Since month-to-month customers churn 15x more than two-year customers, and nearly half of all churn happens in year one, offer a meaningful discount or perk (e.g., a free month, waived installation fee) for new customers who commit to a one- or two-year contract at signup — when they're most likely to still be on a flexible plan.
2. **Investigate the fiber optic customer experience.** Fiber customers churn more than twice as often as DSL customers despite paying more. The business should audit fiber pricing versus competitors and reliability/support metrics for fiber customers specifically, since this looks like a product-level problem rather than a general churn tendency.
3. **Bundle a free trial of Tech Support and Online Security into new sign-ups.** Customers with these add-ons churn at roughly a third of the rate of those without. Offering a 3-6 month free trial (with an easy opt-out) could build the habit and stickiness these add-ons create, while giving the business a natural upsell moment once the trial ends.
4. **Nudge electronic-check customers toward automatic payment.** With electronic check customers churning at 45% vs ~15% for automatic payment methods, a small one-time incentive (e.g., a bill credit) to switch to autopay could reduce both

payment friction and the churn associated with it.

Appendix — Full Cleaning & Exploratory Analysis Code

This section is for technical readers and reproduces the full data cleaning and analysis behind the findings above.

```
# --- Data loading & cleaning ---
import pandas as pd

df_raw = pd.read_csv('Telco-Customer-Churn.csv')
print("Raw shape:", df_raw.shape)
print("\nMissing values per column:\n", df_raw.isnull().sum().sum())
print("\nDuplicate rows:", df_raw.duplicated().sum())

# TotalCharges is read as text because 11 brand-new customers have a blank value
# instead of 0. Convert to numeric and fill those blanks with 0.
blank_total_charges = df_raw[pd.to_numeric(df_raw['TotalCharges'], errors='coerce').isnull()]
print("\nRows with blank TotalCharges (all have tenure = 0):")
print(blank_total_charges[['tenure', 'MonthlyCharges', 'TotalCharges']])

df_clean = df_raw.copy()
df_clean['TotalCharges'] = pd.to_numeric(df_clean['TotalCharges'], errors='coerce').fillna(0)
df_clean['Churn_flag'] = (df_clean['Churn'] == 'Yes').astype(int)
print("\nOverall churn rate:", df_clean['Churn_flag'].mean())
```

Raw shape: (7043, 21)

Missing values per column:
0

Duplicate rows: 0

Rows with blank TotalCharges (all have tenure = 0):

	tenure	MonthlyCharges	TotalCharges
488	0	52.55	
753	0	20.25	
936	0	80.85	
1082	0	25.75	
1340	0	56.05	
3331	0	19.85	
3826	0	25.35	
4380	0	20.00	
5218	0	19.70	
6670	0	73.35	
6754	0	61.90	

Overall churn rate: 0.2653698707936959

```
# --- Full exploratory breakdown behind each finding ---
print("Churn rate by contract type:")
print(df_clean.groupby('Contract')['Churn_flag'].mean().sort_values(ascending=False), "\n")

print("Churn rate by tenure bucket:")
bins = [0, 12, 24, 48, 60, 100]
labels = ['0-12 mo', '13-24 mo', '25-48 mo', '49-60 mo', '61+ mo']
df_clean['tenure_bucket'] = pd.cut(df_clean['tenure'], bins=bins, labels=labels, include_lowest=True)
print(df_clean.groupby('tenure_bucket', observed=True)['Churn_flag'].mean(), "\n")

print("Churn rate by internet service:")
print(df_clean.groupby('InternetService')['Churn_flag'].mean().sort_values(ascending=False), "\n")

print("Churn rate by tech support:")
print(df_clean.groupby('TechSupport')['Churn_flag'].mean().sort_values(ascending=False), "\n")

print("Churn rate by online security:")
print(df_clean.groupby('OnlineSecurity')['Churn_flag'].mean().sort_values(ascending=False), "\n")

print("Churn rate by payment method:")
print(df_clean.groupby('PaymentMethod')['Churn_flag'].mean().sort_values(ascending=False), "\n")

print("Average monthly charges by churn status:")
print(df_clean.groupby('Churn')['MonthlyCharges'].mean(), "\n")

print("Churn rate by senior citizen status:")
print(df_clean.groupby('SeniorCitizen')['Churn_flag'].mean())
```

Churn rate by contract type:

```
Contract
Month-to-month    0.427097
One year          0.112695
Two year          0.028319
Name: Churn_flag, dtype: float64
```

Churn rate by tenure bucket:

```
tenure_bucket
0-12 mo    0.474382
13-24 mo   0.287109
25-48 mo   0.203890
49-60 mo   0.144231
61+ mo     0.066098
Name: Churn_flag, dtype: float64
```

Churn rate by internet service:

```
InternetService
Fiber optic    0.418928
DSL            0.189591
No             0.074050
Name: Churn_flag, dtype: float64
```

Churn rate by tech support:

```
TechSupport
No            0.416355
Yes           0.151663
No internet service  0.074050
Name: Churn_flag, dtype: float64
```

Churn rate by online security:

```
OnlineSecurity
No            0.417667
Yes           0.146112
No internet service  0.074050
Name: Churn_flag, dtype: float64
```

Churn rate by payment method:

```
PaymentMethod
Electronic check    0.452854
Mailed check        0.191067
Bank transfer (automatic)  0.167098
Credit card (automatic)   0.152431
Name: Churn_flag, dtype: float64
```

Average monthly charges by churn status:

```
Churn
No    61.265124
Yes   74.441332
Name: MonthlyCharges, dtype: float64
```

Churn rate by senior citizen status:

```
SeniorCitizen
0    0.236062
1    0.416813
Name: Churn_flag, dtype: float64
```